



Pregnancy dermatosis

primary care block/dermatology

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Content

- Physiological changes:
- Pigmentation , hair , nail , glands , vascular
- Non physiological changes



Learning outcomes

- 1. To know the physiological changes in skin and its appendages during pregnancy
- 2. To differentiate non physiological changes
- 3 to diagnose and treat the non physiological ones
- 4.To know the fetal and maternal risks



Introduction

- The skin changes in pregnancy can be either physiological (hormonal), changes in pre-existing skin diseases or development of new pregnancy specific dermatoses.
- Along with this, the pre-existing skin conditions may either improve or exacerbate in pregnancy due to immunological changes in pregnancy.
- As cell mediated immunity is depressed during normal pregnancy, this accounts for increased severity and frequency of skin infections as candidiasis, *Pityriasis* *Vulvovaginitis*.
- There are however few inflammatory skin dermatoses which are specific to pregnancy and are seen only in pregnancy.
- Though most of these skin dermatoses are benign and resolve in postpartum period a few can risk fetal life and require antenatal surveillance

period, a few can risk fetal life and require antenatal surveillance



physiological

usually reassurance, occur at most of pregnant ladies, disappear after pregnancy
May require treatment but not that dangerous on mother nor fetus

During pregnancy there are marked changes in sex hormones, the immune system, and the cardiovascular system, and this can lead to changes in the skin. Many changes in the skin, hair, and nails of pregnant women are common and considered normal (physiological).



1. changes in pigmentation

- Linea nigra, darkening of pigmented areas (ie, nipples, areolae, genitals), and generalised increase in pigmentation appears in the first trimester of pregnancy. It affects 90% of pregnant women, particularly those with skin of colour (Fitzpatrick skin types IV-VI). This pigmentation fades after delivery, but usually not completely.
- Melasma ^(Chloasma) develops in the second half of pregnancy in 70% of women, especially those with dark complexions. This usually presents as irregular, sharply margined areas of pigmentation in a symmetrical pattern, either on the forehead and temples, or on the central face. Melasma usually fades completely after pregnancy. Prevention includes strict sun protection.
- Melanocytic naevi ^{hyperpigmentation of already existing dots no other changes} may show transient changes in clinical and dermoscopic appearance, but any signs suggestive of melanoma should be investigated.

better not focus on color but whether they burn (i)
or tan (VI) during sun-exposure
a method to classify skin color from I-VI
I: whitest
VI: darkest

it remains more
than 2-3 month
of postpartum
the max resolve
treatment



2. stria gravidarum *remains post-partum*

Stretch marks (striae) are *since size of belly gets bigger during this time* extremely common in pregnancy, affecting up to 90% of women in the second and third trimesters. They are linear, pink or purple, atrophic bands that develop at right angles to the skin tension lines on the abdomen, breasts, thighs, and buttocks.





→ stria gravidarum



→ melasma



→ melanocytic nevus



3. Hair changes

- Many women notice their hair becomes thicker during pregnancy. In the third trimester, the proportion of hair follicles retained in the anagen (growing) phase increases.
- Telogen effluvium post-delivery is a compensatory decrease in hair growth and shedding of hair. Spontaneous recovery back to pre-pregnancy levels usually occurs within 6–12 months. Mild frontoparietal recession (female pattern hair loss) may occur.
- Hypertrichosis (increased hair growth in a non-hormonal pattern) is minor and not uncommon, particularly along the midline suprapubic area. New, soft, fine hairs may disappear around six months post-delivery, but coarse hair typically persists.
high → mostly increase in Vellus hair → fine, smooth, unmedullated hair



4. Nail changes

- Nail changes commonly occur during pregnancy. Increased brittleness, leukonychia, onychoschizia, and ingrown toenails are most common. Increased nail growth, onycholysis, melanonychia, grooving of nails, and subungual hyperkeratosis are also described.
edge of nail grows into surrounding skin
brittle, soft splitting nail
brown/black discoloration of nail
white discoloration of nail
- Benign, uniform, symmetrical hyperpigmentation of multiple nails is reported during pregnancy with fading postpartum, however, irregular pigmentation with cuticle involvement should prompt investigation of possible melanoma.



5. Glandular changes

Miliaria, also known as eczema miliaria, is a common skin condition caused by the blockage of eccrine sweat glands and ducts. This obstruction leads to the backflow of eccrine sweat into the dermis or epidermis, resulting in a rash characterized by sweat-filled vesicles under the skin.

- Eccrine sweat gland activity increases across the body, except on the palms, during pregnancy and may present clinically as [hyperhidrosis](#) and [miliaria](#).
- Sebaceous gland excretion of [sebum](#) tends to increase during pregnancy due to increasing levels of maternal progesterone and androgen in the third trimester.
- Montgomery glands provide lubrication to the nipples and areolae for breastfeeding. They enlarge during pregnancy and appear as papules on the areolae (Montgomery tubercles).



6. Vascular changes

The vascular changes of pregnancy are probably brought about by the general increase in vascularity associated with high oestrogen levels and increased blood volume.

- Non-pitting edema of the legs due to increased hydrostatic pressure can occur in up to 50% of normal pregnancies. Involvement of the face and hands has also been described.
- Varicose veins of the legs, haemorrhoids, and vulvar varicosities are frequent complications of pregnancy.
- Spider telangiectases are very common in pregnancy and disappear post-partum. They appear in areas drained by the superior vena cava which include the face, neck, upper chest, and arms.
- Palmar erythema is common, affecting at least 70% of women with Fitzpatrick skin types I and II, and 30% of women with darker skin types.
- Pyogenic granuloma can present on the fingers, face, gums, or vulvovaginal mucosa

skin tone classification from lightest to darkest

→ as easily bleeding nodules on fingers or face or mucosa of etc parts formed due to vascular changes not often related to pyogenicity

Pyogenic granuloma can present on the fingers, face, gums, or vulvovaginal mucosa in 5% of pregnancies. • Superficial thrombophlebitis and deep vein thrombosis in pregnancy is usually associated with factor V Leiden or prothrombin G20210A gene mutation.

Palmar
Erythema



→ Pilonic
Granuloma



→ Spider Telangiectasia





Non physiological changes

less commonly occur, may or may not disappear after pregnancy, may require treatment

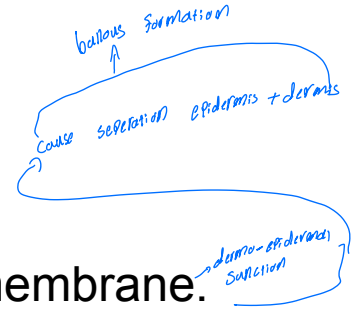
There are four main conditions that occur specifically during pregnancy. These are called pregnancy-specific dermatoses.

- Intrahepatic cholestasis of pregnancy
- Pemphigoid gestationis
↳ blistering (bullous)
- polymorphic eruption of pregnancy
- Atopic eruption of pregnant (prurigo of pregnancy)



1. Pemphigoid gestations

- It is a rare autoimmune pregnancy related blistering skin disorder.
- Usually happens in second or third trimester or in postpartum period.
- Due to specific IgG antibodies against the BP-180 protein in basement membrane.
- Might lead to premature deliveries , small for gestational age or transient blistering in neonate due to transplacental delivery of antibodies in %10.





Clinical presentation

- Initially there are itchy red bumps around the belly button.
- Within days to weeks, the rash spreads to other parts of the body, including the trunk, buttocks, and arms. The face, scalp, palms, soles, and mucous membranes are not usually affected.
- After 2-4 weeks, large, tense blisters filled with clear yellowish fluid may form. Some patients may have no blisters but instead have plaques of inflamed skin.

not affecting mucous membranes (like mouth)

late stage



initial stage





Treatment

- Topical steroid in mild cases
- Systemic steroid in more ^{extensive} ~~expensive~~ cases
- Oral antihistamine
- Azathioprine or cyclosporin
- plasmapheresis
- intravenous immunoglobulin

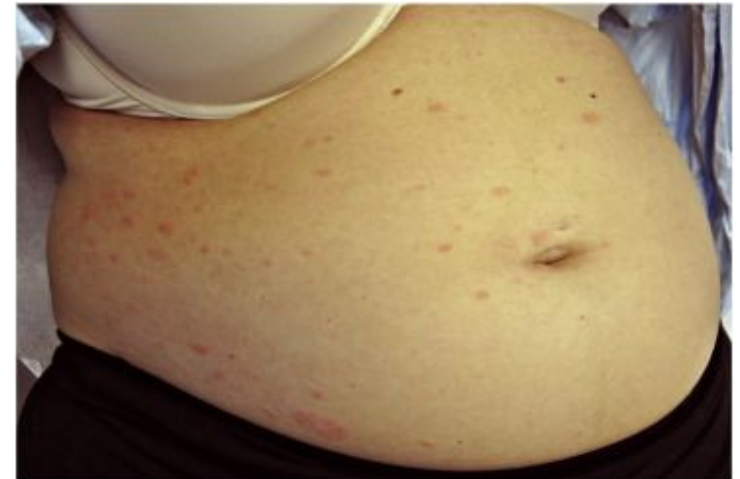
→ very severe

→ despite that are usually contra-indicated in pregnancy but in here we may need to do risk-benefit analyses



2. Prurigo of pregnancy

- It is a benign non specific itchy rash on unknown cause
- Generally appears in second or third trimester
- It can occur in 1 in 300 of pregnancies
- It can disappear spontaneously after pregnancy
- Not associated with any fetal or maternal risk





Clinical features

- typically presents as papular dermatitis. It may resemble [nodular prurigo](#).
- The papules are often grouped.
- They are predominately found on the extensor surfaces of the limbs and sometimes the trunk
- Papules are 0.5–1 cm in size.
- They may or may not be centred on a hair follicle.
- Some lesions have a central crust.
- Pustules may be seen, but blisters (vesicles) are not.
- The papules are often excoriated, ^{due to scratching} due to the itch.



Treatment

- emollients
- benzoyl peroxide *→ if pustule*
- moderate potency steroids
- Oral antihistamine
- phototherapy *→ if resistant patient*



3. polymorphic eruptions of pregnancy

- Itchy bumpy rash that starts in the stretch marks of abdomen during the third trimester. *and inner thighs*
- Also known as PUPP (pruritic urticarial papule and plaques of pregnancy).
- Could be due to the immune reaction elicited by stretching
- More in white women , greater weight , multiple pregnancies , primigravida, male fetus. *→ Twin tripple* *→ first pregnancy*
- Usually disappears 4-6 weeks after delivery without any effect on fetus.





Clinical features

- The rash usually begins during the third trimester, in the last few weeks of pregnancy. About 15% of women with polymorphic eruption of pregnancy report that it actually began immediately after delivery.
- Small pink papules appear first in the stretch marks around the umbilicus, often with a pale halo around each papule.
- The papules coalesce to form large, red, urticated (hive-like) plaques, and occasionally small vesicles are present. The rash spreads from the abdomen to involve the buttocks and thighs, and sometimes the arms and legs. Lesions on or above the breasts are rare.
- Polymorphic eruption of pregnancy is very itchy (pruritic) and patients find it difficult to sleep at night.



Treatment

- emollient
- Topical steroid
- antihistamine *is a must due to sleep disturbance*
may be bad for fetus
- Short course systemic steroid
if severe

4.intrahepatic cholestasis of pregnancy

- Intrahepatic cholestasis of pregnancy, commonly known as cholestasis of pregnancy, is a liver condition that can occur in late pregnancy. *main problem in liver*
- Exact cause is unknown , genetic and hormonal factors may affect bile flow.
- Mostly occurs in third trimester and has 60-70% chances to repeat in next pregnancies.
- Fetal risks : preterm delivery , meconium aspiration , stillbirth *fetus affected in here, perinatal outcomes*
maternal complications
- Maternal risks : clotting problems.



Clinical features

- intense itching mainly in palms and soles without rash.

- jaundice

- Dark urine

- Oily foul smelling stool

- Nausea

- Loss of appetite

in 3rd trimester should be danger sign and investigated





Treatment

- Rest
- emollient
- antihistamine
- Ursodeoxycholic acid
- Termination of pregnancy *→ sadly mother result in this*